Experiment 4

Import and Preprocess Indian Open Datasets Using R

PART B

(PART B: TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no ERP access available)

Roll No.: 60	Name: Avdhut Parish Madhavi
Class : SE-C	Batch : C2
Date of Experiment:	Date of Submission :
Grade :	

Observation: (SUMMARY TABLE IN SHORT FOR EACH SIMULATOR)

INPUT

Kinjal.r*

Source on Save

```
1  getwd()
2  setwd("C:/Users/Admin/Downloads")
3
4  literacy <- read.csv("datafile - datafile.csv",
5                       stringsAsFactors = FALSE)
6
7  literacy
8  head(literacy)
9  tail(literacy)
10 str(literacy)
11 dim(literacy)
12 nrow(literacy)
13 ncol(literacy)
14 names(literacy)
15
```

OUTPUT

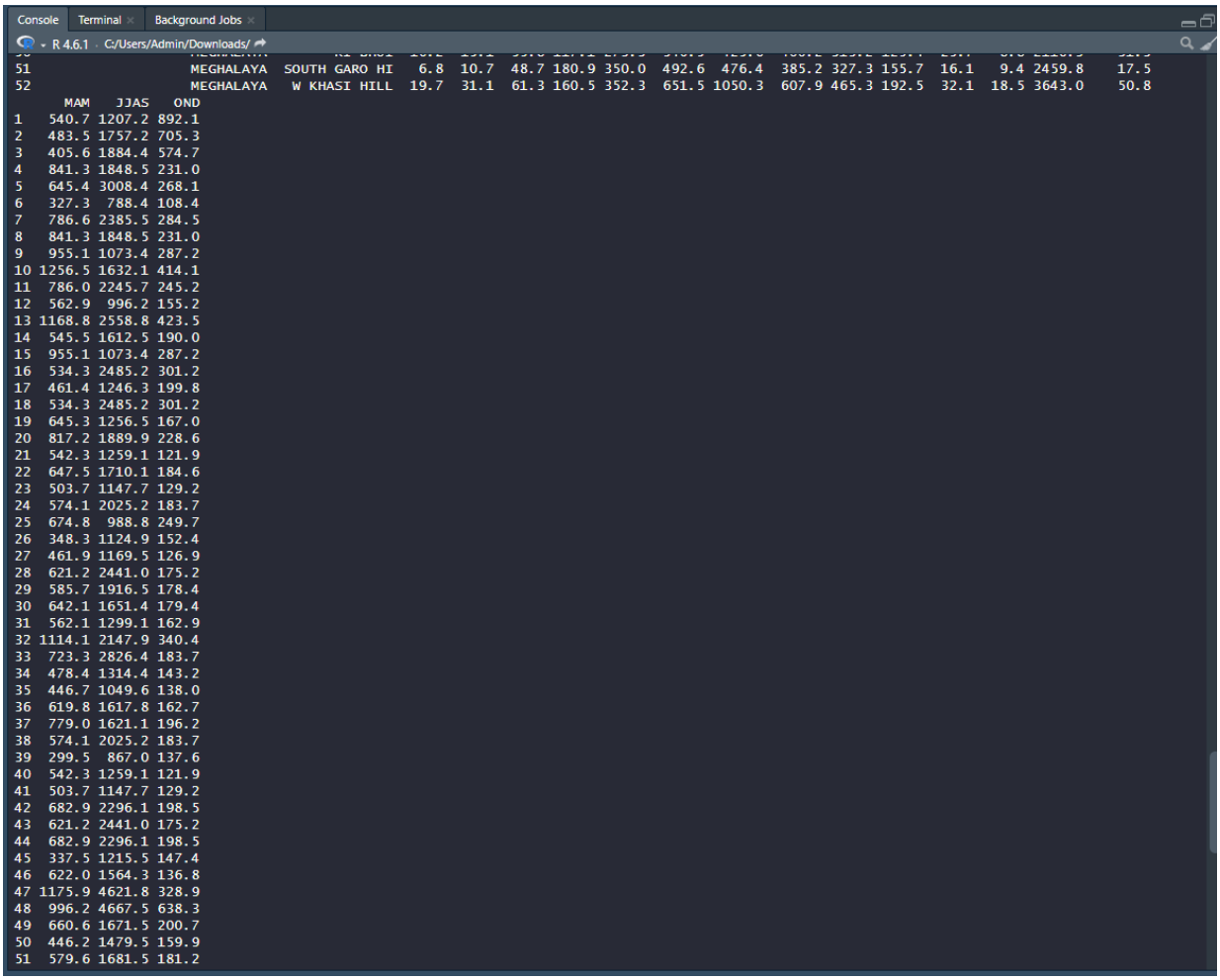
```
> getwd()
[1] "C:/Users/Admin/Documents"
> setwd("C:/Users/Admin/Downloads")
> getwd()
[1] "C:/Users/Admin/Downloads"
> literacy <- read.csv("literacy_rate.csv", stringsAsFactors = FALSE)
```

Console | Terminal | Background Jobs

R 4.6.1 · C:/Users/Admin/Downloads/

```
[249] "wps_rtd.rtd-rhzxyrPshkzB.exe"
> literacy <- read.csv("datafile - datafile.csv",
+                      stringsAsFactors = FALSE)
> literacy
```

	STATE.UT	DISTRICT	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANNUAL	JAN.FEB
1	ANDAMAN And NICOBAR ISLANDS	NICOBAR	107.3	57.9	65.2	117.0	358.5	295.5	285.0	271.9	354.8	326.0	315.2	250.9	2805.2	165.2
2	ANDAMAN And NICOBAR ISLANDS	SOUTH ANDAMAN	43.7	26.0	18.6	90.5	374.4	457.2	421.3	423.1	455.6	301.2	275.8	128.3	3015.7	69.7
3	ANDAMAN And NICOBAR ISLANDS	N & M ANDAMAN	32.7	15.9	8.6	53.4	343.6	503.3	465.4	460.9	454.8	276.1	198.6	100.0	2913.3	48.6
4	ARUNACHAL PRADESH	LOHIT	42.2	80.8	176.4	358.5	306.4	447.0	660.1	427.8	313.6	167.1	34.1	29.8	3043.8	123.0
5	ARUNACHAL PRADESH	EAST SIANG	33.3	79.5	105.9	216.5	323.0	738.3	990.9	711.2	568.0	206.9	29.5	31.7	4034.7	112.8
6	ARUNACHAL PRADESH	SUBANSIRI F.D	28.0	48.3	85.3	101.5	140.5	228.4	217.4	182.8	159.8	75.9	20.9	11.6	1300.4	76.3
7	ARUNACHAL PRADESH	TIRAP	42.2	72.7	141.0	316.9	328.7	614.7	851.9	500.6	418.3	218.7	42.9	22.9	3571.5	114.9
8	ARUNACHAL PRADESH	ANJAW (LOHIT)	42.2	80.8	176.4	358.5	306.4	447.0	660.1	427.8	313.6	167.1	34.1	29.8	3043.8	123.0
9	ARUNACHAL PRADESH	LOWER DIBANG	83.7	153.9	303.5	383.6	268.0	374.2	272.0	160.5	266.7	167.2	64.0	56.0	2553.3	237.6
10	ARUNACHAL PRADESH	CHANGLANG	70.3	170.9	367.9	554.4	334.2	526.2	460.8	291.5	353.6	275.0	64.9	74.2	3543.9	241.2
11	ARUNACHAL PRADESH	PAPUM PARE	33.5	67.8	106.1	226.9	453.0	640.5	609.5	503.4	492.3	214.7	19.2	11.3	3378.2	101.3
12	ARUNACHAL PRADESH	LOW SUBANSIRI	97.5	109.3	92.4	204.3	266.2	284.1	248.9	270.5	192.7	78.5	49.5	27.2	1921.1	206.8
13	ARUNACHAL PRADESH	UPPER SIANG	74.3	176.7	362.6	397.5	408.7	801.9	653.0	417.9	686.0	264.9	86.9	71.7	4402.1	251.0
14	ARUNACHAL PRADESH	WEST SIANG	26.0	66.7	76.8	229.2	239.5	416.6	592.4	312.4	291.1	126.8	33.7	29.5	2440.7	92.7
15	ARUNACHAL PRADESH	DIBANG VALLEY	83.7	153.9	303.5	383.6	268.0	374.2	272.0	160.5	266.7	167.2	64.0	56.0	2553.3	237.6
16	ARUNACHAL PRADESH	WEST KAMENG	35.2	43.5	58.9	134.3	341.1	665.3	749.9	579.1	490.9	233.9	40.3	27.0	3399.4	78.7
17	ARUNACHAL PRADESH	EAST KAMENG	49.0	74.4	96.5	156.9	208.0	345.7	368.5	256.2	275.9	138.2	34.4	27.2	2030.9	123.4
18	ARUNACHAL PRADESH	TAWANG(W KAME	35.2	43.5	58.9	134.3	341.1	665.3	749.9	579.1	490.9	233.9	40.3	27.0	3399.4	78.7
19	ARUNACHAL PRADESH	KURUNG KUMEY	82.7	70.0	128.2	245.7	271.4	292.7	404.0	276.3	283.5	92.3	32.3	42.4	2221.5	152.7
20	ASSAM	CACHAR	13.3	50.2	168.3	262.5	386.4	532.1	526.2	470.8	360.8	182.4	34.8	11.4	2999.2	63.5
21	ASSAM	DARRANG	13.1	21.4	53.5	168.8	320.0	419.7	345.8	272.1	221.5	95.4	17.2	9.3	1957.8	34.5
22	ASSAM	GOALPARA	12.7	20.4	51.1	196.6	399.8	567.8	502.8	334.6	304.9	157.7	21.7	5.2	2575.3	33.1
23	ASSAM	KAMRUP	12.0	20.8	58.6	151.7	293.4	365.5	345.1	248.7	188.4	106.6	15.1	7.5	1813.4	32.8
24	ASSAM	LAKHIMPUR	27.7	48.6	76.7	165.5	331.9	528.3	605.2	467.6	424.1	140.3	23.0	20.4	2859.3	76.3
25	ASSAM	NORTH CACHAR	16.7	47.5	158.9	207.9	308.0	328.1	270.3	201.3	189.1	196.4	42.1	11.2	1977.5	64.2
26	ASSAM	NAGAON	12.0	22.5	48.1	128.9	171.3	285.9	326.3	294.1	218.6	120.0	21.6	10.8	1660.1	34.5
27	ASSAM	SIVASAGAR	20.1	33.2	78.3	126.5	257.1	255.5	374.8	342.3	196.9	96.3	20.3	10.3	1811.6	53.3
28	ASSAM	BARPETA	10.3	26.9	54.0	175.7	391.5	694.3	757.3	527.3	462.1	142.1	20.4	12.7	3274.6	37.2
29	ASSAM	DHUBRI	10.3	11.7	46.6	147.5	391.6	603.0	554.7	418.7	340.1	155.1	19.2	4.1	2702.6	22.0
30	ASSAM	DIBRUGARH	30.6	53.1	119.8	229.9	292.4	401.0	519.4	405.1	325.9	136.9	24.0	18.5	2556.6	83.7
31	ASSAM	JORHAT	22.2	36.7	80.1	204.6	277.4	288.8	390.7	346.9	272.7	121.7	25.6	15.6	2083.0	58.9
32	ASSAM	KARIMGANJ	13.2	35.2	169.7	340.4	604.0	645.2	646.0	438.0	418.7	240.4	86.8	13.2	3650.8	48.4
33	ASSAM	KOKRAJHAR	10.9	27.9	45.8	216.4	461.1	822.2	864.2	677.1	462.9	159.5	18.1	6.1	3772.2	38.8
34	ASSAM	SHONITPUR	19.4	23.1	50.0	144.3	284.1	363.1	384.2	337.9	229.2	109.7	20.9	12.6	1978.5	42.5
35	ASSAM	GOLAGHAT	16.9	29.5	65.2	135.9	245.6	254.2	314.7	271.4	209.3	103.7	19.6	14.7	1680.7	46.4
36	ASSAM	TINSUKIA	26.6	58.4	131.2	215.9	272.7	388.3	515.3	387.2	327.0	118.7	25.1	18.9	2485.3	85.0
37	ASSAM	HAILAKANDI	7.3	42.0	123.8	239.0	416.2	470.4	426.8	406.9	317.0	150.6	36.5	9.1	2645.6	49.3
38	ASSAM	DHEMAJI(LAKHI	27.7	48.6	76.7	165.5	331.9	528.3	605.2	467.6	424.1	140.3	23.0	20.4	2859.3	76.3
39	ASSAM	KARBI ANGLONG	12.8	24.1	53.8	106.3	139.4	224.9	238.7	220.5	182.9	100.1	26.3	11.2	1341.0	36.9
40	ASSAM	UDALGURI(DARA	13.1	21.4	53.5	168.8	320.0	419.7	345.8	272.1	221.5	95.4	17.2	9.3	1957.8	34.5
41	ASSAM	KAMRUP METROP	12.0	20.8	58.6	151.7	293.4	365.5	345.1	248.7	188.4	106.6	15.1	7.5	1813.4	32.8
42	ASSAM	CHIRANG(BONGAI	10.0	31.2	57.5	183.4	442.0	626.3	776.6	486.8	406.4	168.2	18.7	11.6	3218.7	41.2
43	ASSAM	BAKSA BARPETA	10.3	26.9	54.0	175.7	391.5	694.3	757.3	527.3	462.1	142.1	20.4	12.7	3274.6	37.2
44	ASSAM	BONGAIGAON	10.0	31.2	57.5	183.4	442.0	626.3	776.6	486.8	406.4	168.2	18.7	11.6	3218.7	41.2
45	ASSAM	MORIGAON	18.7	24.4	51.6	116.8	169.1	310.1	374.8	309.2	221.4	119.3	19.9	8.2	1743.5	43.1
46	ASSAM	NALBARI	14.9	18.6	56.7	184.6	380.7	551.2	470.9	322.0	220.2	110.4	21.6	4.8	2356.6	33.5
47	MEGHALAYA	EAST KHASI HI	15.4	24.1	129.7	312.5	733.7	1476.2	1518.4	1019.4	607.8	277.9	40.3	10.7	6166.1	39.5
48	MEGHALAYA	JAINTIA HILLS	33.8	44.1	115.1	282.3	598.8	1316.1	1591.3	933.8	826.3	517.7	110.9	9.7	6379.9	77.9
49	MEGHALAYA	EAST GARO HIL	5.6	16.0	64.0	244.4	352.2	531.1	465.8	358.9	315.7	154.1	35.8	10.8	2554.4	21.6
50	MEGHALAYA	RI-BHOI	16.2	15.1	55.8	117.1	273.3	340.5	425.6	400.2	313.2	125.4	25.7	8.8	2116.9	31.3




```
51                  MEGHALAYA  SOUTH GARO HI   6.8  10.7  48.7 180.9 350.0  492.6  476.4  385.2 327.3 155.7  16.1   9.4 2459.8    17.5
52                  MEGHALAYA   W KHASI HILL  19.7  31.1  61.3 160.5 352.3  651.5 1050.3  607.9 465.3 192.5  32.1  18.5 3643.0    50.8
      MAM    JJAS   OND
1   540.7 1207.2 892.1
2   483.5 1757.2 705.3
3   405.6 1884.4 574.7
4   841.3 1848.5 231.0
5   645.4 3008.4 268.1
6   327.3  788.4 108.4
7   786.6 2385.5 284.5
8   841.3 1848.5 231.0
9   955.1 1073.4 287.2
10 1256.5 1632.1 414.1
11  786.0 2245.7 245.2
12  562.9  996.2 155.2
13 1168.8 2558.8 423.5
14  545.5 1612.5 190.0
15  955.1 1073.4 287.2
16  534.3 2485.2 301.2
17  461.4 1246.3 199.8
18  534.3 2485.2 301.2
19  645.3 1256.5 167.0
20  817.2 1889.9 228.6
21  542.3 1259.1 121.9
22  647.5 1710.1 184.6
23  503.7 1147.7 129.2
24  574.1 2025.2 183.7
25  674.8  988.8 249.7
26  348.3 1124.9 152.4
27  461.9 1169.5 126.9
28  621.2 2441.0 175.2
29  585.7 1916.5 178.4
30  642.1 1651.4 179.4
31  562.1 1299.1 162.9
32 1114.1 2147.9 340.4
33  723.3 2826.4 183.7
34  478.4 1314.4 143.2
35  446.7 1049.6 138.0
36  619.8 1617.8 162.7
37  779.0 1621.1 196.2
38  574.1 2025.2 183.7
39  299.5  867.0 137.6
40  542.3 1259.1 121.9
41  503.7 1147.7 129.2
42  682.9 2296.1 198.5
43  621.2 2441.0 175.2
44  682.9 2296.1 198.5
45  337.5 1215.5 147.4
46  622.0 1564.3 136.8
47 1175.9 4621.8 328.9
48  996.2 4667.5 638.3
49  660.6 1671.5 200.7
50  446.2 1479.5 159.9
51  579.6 1681.5 181.2
```

```
Console   Terminal   Background Jobs
R 4.6.1 · C:/Users/Admin/Downloads/
51  579.6 1681.5 181.2
52  574.1 2775.0 243.1
 [ reached 'max' / getOption("max.print") -- omitted 588 rows ]
> head(literacy)
                  STATE.UT      DISTRICT   JAN  FEB   MAR   APR   MAY   JUN   JUL   AUG   SEP   OCT   NOV   DEC ANNUAL JAN.FEB   MAM
1 ANDAMAN And NICOBAR ISLANDS       NICOBAR 107.3 57.9  65.2 117.0 358.5 295.5 285.0 271.9 354.8 326.0 315.2 250.9 2805.2   165.2 540.7
2 ANDAMAN And NICOBAR ISLANDS SOUTH ANDAMAN  43.7 26.0  18.6  90.5 374.4 457.2 421.3 423.1 455.6 301.2 275.8 128.3 3015.7    69.7 483.5
3 ANDAMAN And NICOBAR ISLANDS N & M ANDAMAN  32.7 15.9   8.6  53.4 343.6 503.3 465.4 460.9 454.8 276.1 198.6 100.0 2913.3    48.6 405.6
4           ARUNACHAL PRADESH         LOHIT  42.2 80.8 176.4 358.5 306.4 447.0 660.1 427.8 313.6 167.1  34.1  29.8 3043.8   123.0 841.3
5           ARUNACHAL PRADESH    EAST SIANG  33.3 79.5 105.9 216.5 323.0 738.3 990.9 711.2 568.0 206.9  29.5  31.7 4034.7   112.8 645.4
6           ARUNACHAL PRADESH SUBANSIRI F.D  28.0 48.3  85.3 101.5 140.5 228.4 217.4 182.8 159.8  75.9  20.9  11.6 1300.4    76.3 327.3
    JJAS   OND
1 1207.2 892.1
2 1757.2 705.3
3 1884.4 574.7
4 1848.5 231.0
5 3008.4 268.1
6  788.4 108.4
> tail(literacy)
    STATE.UT       DISTRICT  JAN  FEB  MAR   APR   MAY   JUN    JUL   AUG   SEP   OCT   NOV  DEC ANNUAL JAN.FEB   MAM   JJAS   OND
635   KERALA       THRISSUR  3.4  7.5 17.0  79.2 289.1 707.4  772.9 440.1 277.1 296.3 138.1 35.0 3063.1    10.9 385.3 2197.5 469.4
636   KERALA  THIRUVANANTHA 19.3 21.1 35.5 116.7 216.6 338.7  228.9 142.0 161.7 268.0 192.0 62.7 1803.2    40.4 368.8  871.3 522.7
637   KERALA         IDUKKI 13.4 22.1 43.6 150.4 232.6 651.6  788.9 527.3 308.4 343.2 172.9 48.1 3302.5    35.5 426.6 2276.2 564.2
638   KERALA       KASARGOD  2.3  1.0  8.4  46.9 217.6 999.6 1108.5 636.3 263.1 234.9  84.6 18.4 3621.6     3.3 272.9 3007.5 337.9
639   KERALA PATHANAMTHITTA 19.8 45.2 73.9 184.9 294.7 556.9  539.9 352.7 266.2 359.4 213.5 51.3 2958.4    65.0 553.5 1715.7 624.2
640   KERALA        WAYANAD  4.8  8.3 17.5  83.3 174.6 698.1 1110.4 592.9 230.7 213.1  93.6 25.8 3253.1    13.1 275.4 2632.1 332.5
> str(literacy)
'data.frame':   640 obs. of  19 variables:
 $ STATE.UT: chr  "ANDAMAN And NICOBAR ISLANDS" "ANDAMAN And NICOBAR ISLANDS" "ANDAMAN And NICOBAR ISLANDS" "ARUNACHAL PRADESH" ...
 $ DISTRICT: chr  "NICOBAR" "SOUTH ANDAMAN" "N & M ANDAMAN" "LOHIT" ...
 $ JAN     : num  107.3 43.7 32.7 42.2 33.3 ...
 $ FEB     : num  57.9 26 15.9 80.8 79.5 ...
 $ MAR     : num  65.2 18.6 8.6 176.4 105.9 ...
 $ APR     : num  117 90.5 53.4 358.5 216.5 ...
 $ MAY     : num  358 374 344 306 323 ...
 $ JUN     : num  296 457 503 447 738 ...
 $ JUL     : num  285 421 465 660 991 ...
 $ AUG     : num  272 423 461 428 711 ...
 $ SEP     : num  355 456 455 314 568 ...
 $ OCT     : num  326 301 276 167 207 ...
 $ NOV     : num  315.2 275.8 198.6 34.1 29.5 ...
 $ DEC     : num  250.9 128.3 100 29.8 31.7 ...
 $ ANNUAL  : num  2805 3016 2913 3044 4035 ...
 $ JAN.FEB : num  165.2 69.7 48.6 123 112.8 ...
 $ MAM     : num  541 484 406 841 645 ...
 $ JJAS    : num  1207 1757 1884 1848 3008 ...
 $ OND     : num  892 705 575 231 268 ...
> dim(literacy)
[1] 640  19
> nrow(literacy)
[1] 640
> ncol(literacy)
[1] 19
```

```
> names(literacy)
 [1] "STATE.UT" "DISTRICT" "JAN"      "FEB"      "MAR"      "APR"      "MAY"      "JUN"      "JUL"      "AUG"      "SEP"      "OCT"
[13] "NOV"      "DEC"      "ANNUAL"   "JAN.FEB"  "MAM"      "JJAS"     "OND"
>
```

Result : The literacy dataset was successfully imported into R. The head() and tail() functions displayed the first and last six records, str() showed the dataset structure and data types, dim(), nrow(), and ncol() displayed the number of rows and columns, and names() listed all column names.

.

Conclusion: The experiment was completed successfully. The CSV file was imported into R, and the dataset was explored using basic R functions to view records, identify its structure, dimensions, and column names.

Questions of Curiosity

Experiment No 4

* Question of curosity

Q1. Data preprocessing is the process of cleaning, organizing and transforming raw data before performing analysis.

Q2 Preprocessing is necessary to remove errors handles missing value & make the data suitable for accurate analysis.

Q3. The read csv () Function is used to import a csv file in R
example: data <- read.csv ("data.csv")

Q4 head() displays the first few row of dataset tail() display the last few rows of dataset.

Q5. the str() Function shows the structure of a dataset including the number of observation variables, data types and sample values.

Viva Questions

1. What is data preprocessing?
2. Why is preprocessing necessary before data analysis?
3. Which function is used to import a CSV file in R?
4. What is the purpose of head() and tail()?
5. What information does str() provide?
6. How do you identify missing values in R?
7. What is the use of na.omit()?
8. How do you remove duplicate records?
9. How can you rename a column in a data frame?
10. How do you convert a character column into numeric?
11. What is the difference between summary() and str()?
12. Why should cleaned data be saved separately?
13. What is the purpose of the subset() function?
14. How do you sort a dataset in ascending and descending order?
15. Name any four sources of Indian open datasets.